

LAB 5

CONFIGURATION OF STATIC ROUTES AND DEFAULT ROUTES

Objectives

- To learn the principles of Static Routing and Default Routing used for manual path selection in a network.
- To set up a network topology in Cisco Packet Tracer utilizing Static and Default routing methods.
- To enable communication between multiple networks using Static and Default routing configurations.
- To observe and evaluate packet flow across networks in order to confirm routing behavior and decisions.

THEORY

1. Static Routing

Static routing refers to the manual configuration of routing entries in a router's routing table. In this approach, the network administrator defines the precise route that data packets must take to reach a specific destination network.

Operation: Static routes are set using the command:

```
ip route [destination_network] [subnet_mask] [next_hop_address]
```

The router relies on this entry to forward packets. If the specified link fails, the administrator must manually adjust or update the route.

Application: Static routing is commonly used in smaller networks with straightforward and stable topologies. It is also useful for security purposes, such as limiting external visibility of certain network segments.

2. Default Routing

Default routing is a type of static routing that functions as the *gateway of last resort*. With this method, when a router does not find a specific destination network listed in its routing table, it forwards the packet to a predefined next-hop router.

Operation: A default route is configured using the command:

```
ip route 0.0.0.0 0.0.0.0 [next_hop_address]
```

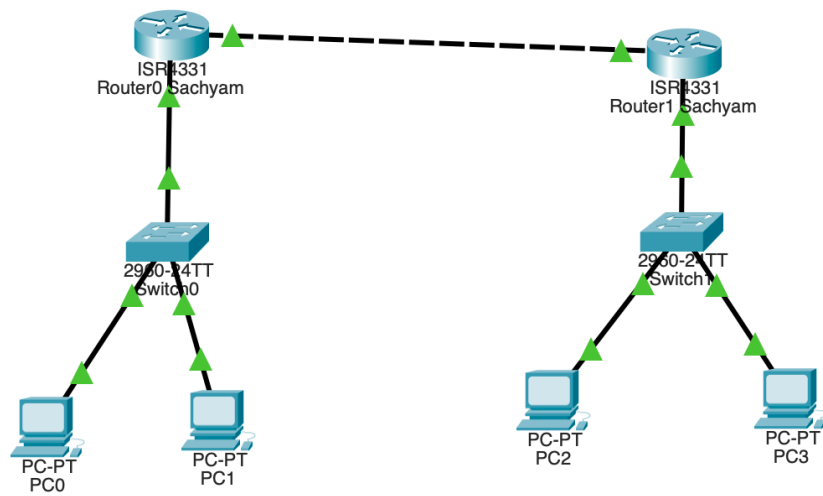
This command directs the router to send any traffic with an unknown destination to the specified neighboring router.

Application: Default routing is primarily used in stub networks (networks that have only one exit path) or when connecting a local network to the Internet through an ISP. It simplifies routing management by minimizing the number of entries required in the routing table.

NETWORK TOPOLOGY

The illustrated network topology includes two routers—Router0 and Router1 Sachyam—connected through a serial link that enables communication between two separate networks.

- On the left side, Router0 Sachyam is linked to Switch0, which connects PC0 and PC1, creating the first LAN segment.
- On the right side, Router1 Sachyam is connected to Switch1, which links PC2 and PC3, forming the second LAN segment.
- The serial link between the two routers allows data packets to move between both LANs, ensuring complete inter-network communication.
- The connection between the routers is established using a cross-cable.



Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
Router0 Sachyam	GigabitEthernet0/0	10.0.0.1	255.0.0.0	N/A
Router0 Sachyam	GigabitEthernet0/1	192.168.1.1	255.255.255.0	N/A
Router1 Sachyam	GigabitEthernet0/0	10.0.0.2	255.0.0.0	N/A
Router1 Sachyam	GigabitEthernet0/1	192.168.2.1	255.255.255.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

Routing Configuration

Syntax :

ip route <destination_network> <subnet_mask> <next_hop_ip>

Device	Routing Type	Command / Route
Router0 Sachyam	Static Route	ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router1 Sachyam	Static Route	ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router0 Sachyam	Default Route	ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router1 Sachyam	Default Route	ip route 0.0.0.0 0.0.0.0 10.0.0.1

RESULT

The network setup was successfully completed. Connectivity was verified by sending a ping from PC2 (192.168.2.2) in the second network to PC0 (192.168.1.2) in the first network. The packets were sent and received without any loss, confirming that the static routing configuration was correct. A similar test was performed for the default routing, which also operated as intended.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: static routing

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=34ms TTL=128
Reply from 192.168.1.3: bytes=32 time=20ms TTL=128
Reply from 192.168.1.3: bytes=32 time=21ms TTL=128
Reply from 192.168.1.3: bytes=32 time=19ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 34ms, Average = 23ms
```

Fig: Dynamic Routing

DISCUSSION

In this experiment, both static and default routing were implemented to establish communication between two distinct LANs connected via routers. Static routing allowed packets to travel along a specifically defined path, whereas default routing served as a fallback route for destinations not listed in the routing table. Ping tests verified that both routing approaches operated correctly, enabling smooth data transfer between the networks. This highlights how different routing strategies influence network connectivity and overall performance.

CONCLUSION

The network topology was successfully created using Cisco Packet Tracer, and the functionality of both static and default routing was confirmed through connectivity tests. Static routing offered detailed control over packet forwarding paths, while default routing simplified network configuration where only one exit route was available. In summary, the experiment emphasized the critical role of routing methods in maintaining dependable communication across networks.